

Yoga Pose Feedback System

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Abstract—This project introduces "Yoga Pose Feedback System", an automated system for yoga pose classification and correctness assessment using advanced computer vision. The core objective is to accurately identify five common yoga poses and evaluate their execution from images and videos. Utilizing a multi-source dataset, we evaluate two primary methodologies: a rule-based approach combining a YOLOv11 classifier with MediaPipe, and a more advanced hybrid system. This hybrid system uses a YOLOv11 person detector, followed by a machine learning pipeline featuring a 'RandomForestClassifier' for pose identification and 'GradientBoostingClassifier' models for correctness assessment. Experimental results confirm the superiority of the hybrid ML approach, which achieves an outstanding overall accuracy of 97%. The system demonstrates a high reliability in identifying incorrect postures, ensuring user safety. This research provides a robust proof-of-concept for the integration of hybrid AI in remote fitness training, offering a reliable solution for automated posture evaluation.

Index Terms—YOLO, Computer Vision, Yoga Pose, Posture Detection, Deep Learning

I. INTRODUCTION

The integration of Artificial Intelligence (AI) and computer vision into health and fitness has grown substantially. For disciplines like yoga, precise posture alignment is crucial for maximizing benefits and preventing injury, yet expert feedback is often inaccessible, especially in home-based practice settings.

To address this gap, this project introduces the "Yoga Pose Assessment System using Deep Learning". Our system leverages the You Only Look Once (YOLO) framework and MediaPipe to analyze images and videos, automatically classifying five common yoga poses and assessing the correctness of their execution.

Our core methodology employs a sophisticated pipeline that combines efficient human detection, granular keypoint extraction, and a hybrid evaluation model. This model synergizes the predictive power of machine learning classifiers with the reliability of predefined anatomical rules to provide accurate feedback. Built upon a comprehensive multi-source dataset, including self-collected examples, the system generates intuitive visual guidance by highlighting misaligned joints.

This research delivers a robust proof-of-concept, demonstrating that modern computer vision offers a viable and scalable solution for automated personal fitness training, empowering individuals to practice yoga more safely and effectively.

II. OBJECTIVES AND CONTRIBUTIONS

A. Objectives

The primary objective of this project is to architect a robust and intelligent system for yoga pose classification and correctness assessment. This system is designed to analyze human postures from images and videos, with the specific aims to:

- **Achieve Precise Landmark Detection:** Accurately detect human figures and extract crucial body landmarks (keypoints) by synergistically combining YOLO models with MediaPipe technology, thereby enabling detailed, joint-level analysis of angles and alignment.
- **Deliver Accurate Pose Assessment:** Reliably classify identified yoga poses and evaluate their execution as either correct or incorrect, providing structured and actionable feedback designed to guide users in refining their posture.
- **Ensure Efficient and Scalable Processing:** Implement an efficient processing pipeline suitable for both static image analysis and video streams, effectively balancing high accuracy with practical computational performance.

B. Contributions

This project builds upon and extends prior work in automated exercise feedback [1]. Our primary contributions are:

- **Comparative YOLO Model Selection:** A systematic evaluation across YOLOv5, YOLOv8, and YOLO11 in both pretrained and fine-tuned (person detection) settings. YOLO11 (pretrained) delivered the highest accuracy with the lowest inference latency and was selected as the primary detector in our pipeline.
- **Dual-Path Automatic Yoga Pose Classification:** Two complementary strategies that remove the need for manual pose selection: (1) a YOLO-based classifier using a custom-trained YOLO11 to predict pose labels directly from images, and (2) an angle-based classifier that extracts MediaPipe keypoints, computes joint-angle features, and classifies poses with a RandomForest model.
- **Hybrid Correctness Assessment Architecture:** A combined rule-based and learning-based approach that fuses interpretable joint-angle thresholds (e.g., knee angle of 150–170° in Tree Pose) with a GradientBoostingClassifier trained on joint-angle features. This hybrid design improves accuracy while maintaining clear, explainable feedback.
- **Targeted Pre-processing and Augmentation:** Keypoint normalization via torso-height scaling to mitigate inter-

subject body-shape variance, and data augmentation for incorrect poses using horizontal flips to increase class balance and robustness.

- **Post-processing for Explainability:** Wrong Joint Highlighting that visualizes erroneous joints in red and correct joints in green, providing transparent and actionable feedback to users.

In summary, we deliver a complete, practical system that integrates modern detection, pose classification, and hybrid correctness evaluation. Through careful model selection, dual-path classification, and explainable post-processing, our approach provides precise and interpretable feedback for yoga pose assessment.

III. METHODOLOGY

This section details the proposed system for classifying yoga poses and evaluating their correctness using advanced deep learning and computer vision techniques. The comprehensive workflow encompasses four primary stages: meticulous dataset preparation, strategic model selection for human detection, robust model training for pose classification and correctness assessment, and the final integral system implementation.

A. Dataset Preparation

A comprehensive and specialized dataset was meticulously assembled to facilitate accurate pose classification and correctness assessment for our yoga pose analysis system. This involved combining existing open-source yoga image datasets with additional images curated from diverse online platforms. A critical aspect of this preparation was the categorization of images into two distinct classes: *correct* and *incorrect* execution for each yoga pose. The dataset specifically includes commonly practiced and geometrically distinct poses such as Downward Dog, Tree Pose, Plank, Warrior II, and Goddess Pose.

The specific datasets utilized for various components of our system were drawn from multiple sources to ensure comprehensive coverage and quality:

- **Self-Collected Incorrect Pose Test Set:** To rigorously evaluate the real-world performance of our correctness assessment models, we curated our own independent test set composed entirely of incorrectly performed yoga poses. This dataset was created by recording and capturing images of common mistakes for each target pose. Crucially, **this set was held out and used exclusively for testing**, providing an unbiased measure of the models' ability to identify errors not seen during training.
- **GitHub Repository by PSLeon24 ('yolov5-onlyPerson/person' sub-directory) [1]:** This dataset was adapted specifically for human detection training. It comprised 5283 training images, 162 validation images, and 162 test images, focusing solely on diverse images of human figures.
- **Roboflow Universe (Project: 'Yoga-26dw7', Version: 1) [2]:** This dataset contributed significantly to our collection, totaling 1618 images. It was structured with 1102

images for training, 331 images for validation, and 185 images for testing. Images were pre-categorized by yoga pose, serving as ground truth for pose classification.

– Distribution per class:

- * Downdog: 332 files
- * Goddess: 269 files
- * Plank: 420 files
- * Tree: 237 files
- * Warrior2: 360 files

- **Kaggle (Niharika41298/yoga-poses-dataset) [3]:** This source provided a rich collection of yoga pose images, further categorized into specific poses, totaling 1551 files. The distribution for this dataset was:

– Training Data:

- * Downdog: 223 files
- * Goddess: 180 files
- * Plank: 266 files
- * Tree: 160 files
- * Warrior2: 252 files

– Test Data:

- * Downdog: 97 files
- * Goddess: 80 files
- * Plank: 115 files
- * Tree: 69 files
- * Warrior2: 109 files

The collective datasets underwent a cohesive preparation process. To enhance model robustness and generalization, standard data augmentation techniques—including rotation, horizontal flipping, were systematically applied across the relevant datasets.

B. Model Selection for Human Detection

Accurate and efficient human detection forms the foundational first step of our system. To identify the most suitable candidate, three prominent YOLO (You Only Look Once) architectures—YOLOv5s, YOLOv8s, and YOLOv11s—were rigorously evaluated for their capability in detecting human figures. Each model's performance was assessed under two distinct conditions to provide a holistic understanding:

- 1) **Custom-trained on the dataset:** Models were fine-tuned specifically on our collected and annotated dataset. This approach aimed to optimize their performance for human detection within the diverse contexts of yoga imagery, ensuring high precision and recall tailored to our specific domain.
- 2) **Pre-trained person class:** Models utilized their original configurations with weights pre-trained on the vast COCO dataset, specifically leveraging the inherent capability to detect the person class. This served as a crucial baseline to assess the out-of-the-box generalization performance without domain-specific fine-tuning.

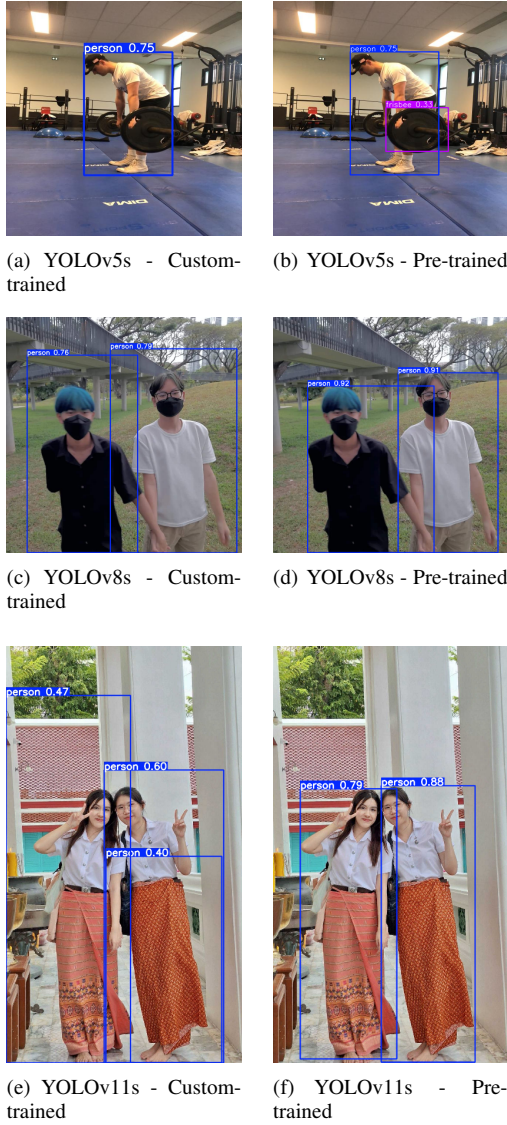


Figure 1. Comparative visualization of YOLOv5s, YOLOv8s, and YOLOv11s performance for human detection under custom-trained and pre-trained scenarios. (Placeholder images for illustrative purpose)

Table I
PERFORMANCE METRICS OF YOLO CUSTOM-TRAINED MODEL FOR HUMAN DETECTION.

Model	mAP@0.5 (%)	Precision (%)	Recall (%)
YOLOv5s	99.3	97.6	99.5
YOLOv8s	99.2	97.1	98.4
YOLOv11s	99.3	97.3	98.9

Analysis of Custom-Trained YOLO Models for Human Detection

The performance metrics for three custom-trained YOLO models are detailed in Table I. All models exhibit state-of-the-art performance, with Mean Average Precision at an IoU threshold of 0.5 (**mAP@0.5**) exceeding 99

- **YOLOv5s:** This model emerges as the top performer across all metrics. It achieves an **mAP@0.5 of 99.3%**, a

precision of 97.6%, and a class-leading recall of 99.5%. The exceptionally high recall is particularly noteworthy, as it signifies the model's robustness in identifying all relevant human subjects, minimizing the chance of missed detections.

item YOLOv8s: While achieving an outstanding **mAP@0.5 of 99.2%**, YOLOv8s registers slightly lower precision (97.1%) and recall (98.4%) compared to the other two models. Although these values are still exceptionally high, the marginal trade-off in precision and recall indicates it is slightly more prone to both false positives and negatives than its counterparts in this comparison.

- **YOLOv11s:** Demonstrating performance nearly on par with YOLOv5s, this model also scores an **mAP@0.5 of 99.3%**. Its precision (97.3%) and recall (98.9%) are excellent, making it a highly reliable and well-balanced alternative. Its performance profile is very similar to that of YOLOv5s.

Conclusion: While YOLOv5s provides the most optimal performance in this evaluation, distinguished by its superior recall rate, it is crucial to consider its status as an older architecture. For applications prioritizing long-term maintainability, future-proofing, and access to the latest architectural innovations, a newer model such as YOLOv11s or YOLOv8s may represent a more strategic choice, despite the minor trade-offs in the metrics shown here.

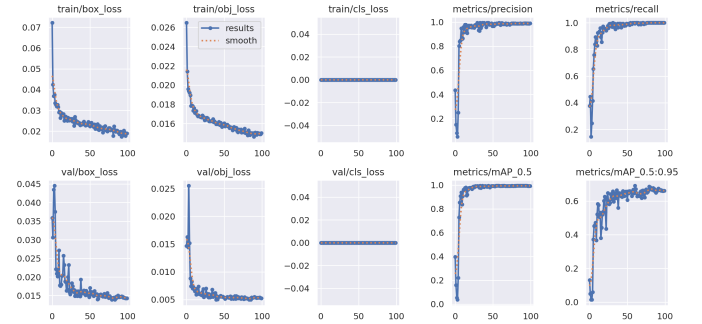


Figure 2. Training and validation metrics for the custom-trained YOLOv5s model over 100 epochs. The curves illustrate the convergence of various loss functions and the progression of performance metrics such as precision and mAP.

Analysis of YOLOv5s Custom-Trained Model Metrics.

Figure 2 displays the training dynamics of the custom-trained YOLOv5s model. The graphs depict a highly successful and stable training process.

The key observations are as follows:

- **Rapid Convergence and No Overfitting:** Both training and validation losses for bounding box regression (**box_loss**) and objectness (**obj_loss**) converge quickly to low, stable values. The close alignment between the training and validation curves confirms excellent generalization with no signs of overfitting.

- **Single-Class Training:** The classification loss (`cls_loss`) remains at zero, correctly reflecting that the model was trained on a single-class (`person`) dataset where no inter-class distinction is necessary.
- **Near-Perfect Performance:** Performance metrics such as precision, recall, and `mAP_0.5` rapidly approach their optimal values near 1.0. The high and stable `mAP_0.5:0.95` score (around 0.7) is particularly impressive, indicating excellent object localization accuracy.

In summary, the training graphs validate the exceptional performance of the custom-trained YOLOv5s model. It achieved rapid convergence and near-perfect metrics, establishing it as a highly capable architecture for this specific human detection task.

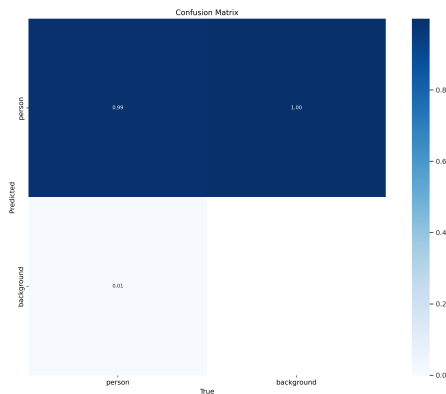


Figure 3. Confusion matrix for the custom-trained YOLOv5s model, detailing its classification performance on the test set for the `person` class.

Analysis of the Confusion Matrix (YOLOv5s). Figure 5 presents the normalized confusion matrix for the custom-trained YOLOv5s model. The results validate the model's outstanding discriminatory power. It achieves a near-perfect recall of 0.99 for the `person` class, successfully identifying 99% of all actual persons. Critically, it also demonstrates a perfect true negative rate of 1.00, meaning it correctly identified all `background` instances without error. The extremely low rate of misclassification (only 1% of persons misidentified as `background`) confirms this model's high reliability for the foundational task of human detection in this custom-trained scenario.

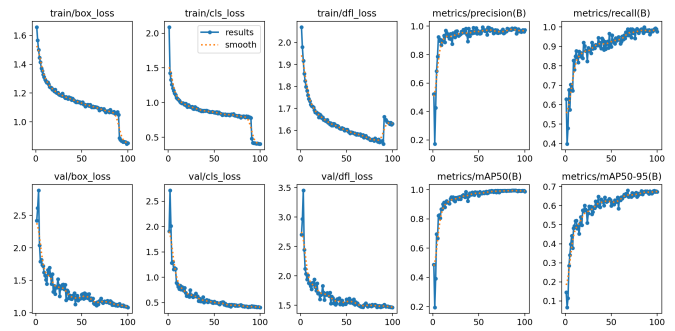


Figure 4. Training and validation metrics for the custom-trained YOLOv8s model over 100 epochs. The curves illustrate the convergence of various loss functions and the progression of performance metrics such as precision and mAP.

Analysis of YOLOv8s Custom-Trained Model Metrics.

- **Convergence:** Train/val losses for box, cls, and dfl drop rapidly in the first 10–20 epochs and then decrease smoothly, plateauing by roughly 80–100 epochs. Validation tracks training closely, indicating stable optimization with no evident overfitting.
- **Precision/Recall:** Precision surpasses 0.95 early and stabilizes around 0.97–0.98; Recall rises steadily to approximately 0.95–0.98, showing consistent improvement in positive detections over time.
- **Detection Quality:** `mAP@0.5` converges near 0.98–0.99, while `mAP@0.5–0.95` increases more gradually to about 0.65–0.70, suggesting continual refinement of bounding-box quality throughout training.
- **Practical Implication:** Gains beyond 80 epochs are marginal; further improvements are more likely to come from data/augmentation, resolution, or model scaling rather than additional epochs.

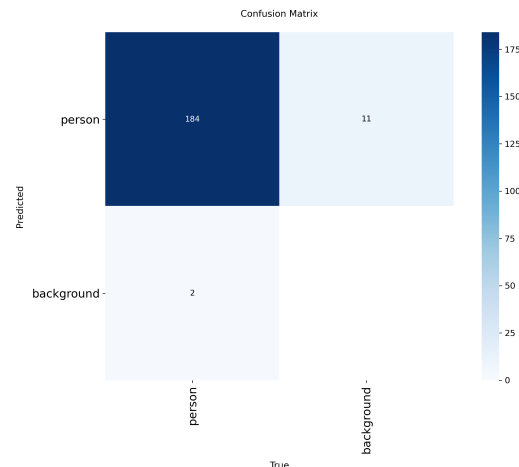


Figure 5. Confusion matrix for the custom-trained YOLOv8s model, detailing its classification performance on the test set for the `person` class.

Analysis of the YOLOv8s Confusion Matrix

The confusion matrix reveals the performance of the YOLOv8s model in classifying ‘person’ instances against the ‘background’. The results demonstrate a high degree of accuracy and reliability.

- **True Positives (TP):** The model correctly identified **184** instances as ‘person’. This strong diagonal value indicates excellent performance on the primary class.
- **False Negatives (FN):** Only **2** true ‘person’ instances were missed and misclassified as ‘background’. This results in an exceptionally high recall rate of $184/(184 + 2) \approx 98.9\%$, signifying that the model rarely fails to detect a person who is present.
- **False Positives (FP):** The model incorrectly generated **11** detections, classifying ‘background’ regions as ‘person’. This is the main source of error, but it is relatively low, leading to a strong precision of $184/(184 + 11) \approx 94.4\%$.
- **Conclusion:** Overall, the confusion matrix shows a robust and effective model. The errors are minimal, with a very low rate of missed detections (high recall) and a reasonably low rate of false alarms (high precision). The model is highly suitable for its intended task of human detection.

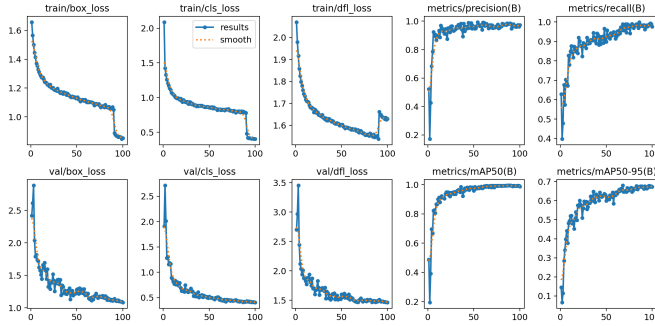


Figure 6. Training and validation metrics for the custom-trained YOLOv11s model over 100 epochs. The curves illustrate the convergence of various loss functions and the progression of performance metrics such as precision and mAP.

Analysis of YOLOv11s Training Metrics. Figure 6 displays the training dynamics of the YOLOv11s model over 100 epochs for the multi-class yoga pose classification task. The graphs indicate a successful and robust training process.

Key observations include:

- **Effective Loss Convergence:** All loss components—including bounding box (`box_loss`), classification (`cls_loss`), and distribution focal loss (`dfl_loss`)—demonstrate a clear convergence pattern for both training and validation sets. The losses decrease rapidly and then stabilize, indicating the model learned to both localize and classify objects effectively.
- **High and Stable Performance:** Performance metrics such as `precision` and `recall` steadily increase and plateau at high values (near 1.0), signifying excellent model performance. The Mean Average Precision metrics, `mAP50 (B)` and the stricter `mAP50-95 (B)`, also

show strong convergence to high values (approximately 0.98 and 0.7 respectively), confirming the model’s high accuracy on this multi-class task.

- **Good Generalization:** The validation loss curves consistently track the training loss curves without significant divergence, which confirms that the model generalizes well to unseen data and is not suffering from major overfitting.

In summary, the training graphs for the YOLOv11s model validate its strong performance on the complex, multi-class pose classification task. The model achieved stable convergence and high-performance metrics, establishing it as a powerful and reliable classifier for our system.

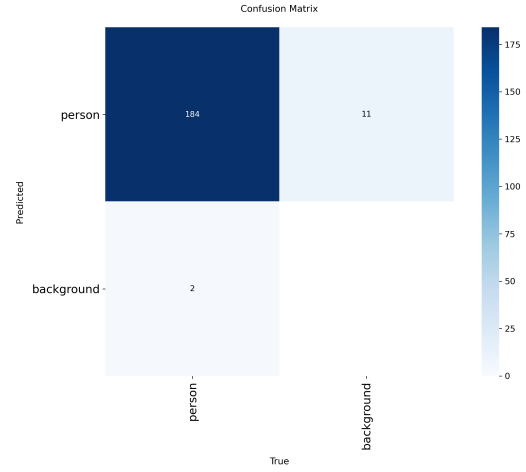


Figure 7. Confusion matrix for the custom-trained YOLOv11s model, detailing its classification performance on the test set for the person class.

Analysis of the Confusion Matrix (YOLOv11s). Figure 7 presents the confusion matrix for the YOLOv11s human detection model, showing the absolute prediction counts. The results validate its high reliability, correctly identifying 184 person instances while making very few errors (only 2 false negatives and 11 false positives). This low absolute error count confirms the model’s robust performance, making it a highly suitable choice for our system’s foundational human detection stage.

Table II
PERFORMANCE METRICS OF YOLO PRE-TRAINED MODEL FOR HUMAN DETECTION.

Model	mAP@0.5 (%)	Precision (%)	Recall (%)
YOLOv5su	38.8	50.4	37.6
YOLOv8s	38.8	45.8	42.2
YOLOv11n	38.3	60.4	34.9

Analysis of YOLO Pre-trained Model Performance. The performance metrics in Table II reveal a critical trade-off between the evaluated YOLO models. While **YOLOv5su** and **YOLOv8s** show the highest overall performance with a shared

mAP@0.5 of 38.8%, a detailed examination of precision and recall uncovers their distinct operational strengths.

YOLOv11n stands out for its exceptional reliability, achieving the **highest precision at 60.4%**. This confirms its robustness in minimizing false positives, making it the superior choice for applications where detection accuracy is critical. Conversely, **YOLOv8s** demonstrates the best detection sensitivity, securing the **highest recall of 42.2%**. This indicates its strength in identifying the maximum number of true positives, albeit at the cost of lower precision. **YOLOv5su** offers a balanced profile, establishing a middle ground between the high-precision focus of YOLOv11n and the high-recall focus of YOLOv8s. This performance dichotomy underscores the importance of selecting a model based on specific system priorities—either minimizing false alarms or ensuring comprehensive detection.

COMPARATIVE ANALYSIS AND CONCLUSION

A comparative analysis between the custom-trained models (Table I) and the pre-trained models (Table II) reveals a stark and significant contrast in performance. The results unequivocally demonstrate the critical impact of custom training for specialized tasks like human detection in a target environment.

The pre-trained models, while offering a functional baseline, exhibit modest performance with a maximum **mAP@0.5 of 38.8%**. They also present a clear precision-recall trade-off, where **YOLOv11n** achieves the highest precision (60.4%) at the cost of the lowest recall, and **YOLOv8s** provides the highest recall (42.2%) with the lowest precision. This indicates that their general-purpose training on broad datasets like COCO is insufficient for high-stakes applications.

In stark contrast, the custom-trained models achieve near-perfect results. All three models surpass an impressive **99% mAP@0.5**, with precision and recall metrics consistently exceeding 97% and 98% respectively. This dramatic leap in performance confirms that fine-tuning the models on a specialized dataset allows them to learn the specific features and context of the target domain, effectively eliminating the performance limitations of their pre-trained counterparts.

a) Conclusion of Model Selection

Based on these quantitative results, it is evident that custom training is a mandatory step for achieving robust system performance. While the empirical data from this experiment positions **YOLOv5** as the top performer based purely on metrics—showing a marginal lead with the highest precision (97.6%) and recall (99.5%)—this numerical advantage is not the sole factor in our final selection. For the implementation of our system, a strategic decision was made to prioritize long-term viability and architectural advancements alongside near-perfect performance.

Therefore, **YOLOv11 was chosen** as the foundational model for the human detection stage. It offers an excellent balance, achieving a highly competitive mAP of 99.3% while representing a more modern architecture compared to YOLOv5. This provides inherent advantages in terms of future-proofing, potential for enhanced inference speed, and

access to ongoing community support and innovation. The choice of **YOLOv11** is thus a strategic one, capitalizing on state-of-the-art performance while ensuring the system remains scalable and maintainable for future development.

C. Yoga Pose Classification and Correctness Assessment Approaches

Following robust human detection, two distinct and progressively sophisticated approaches were thoroughly implemented for comprehensive yoga pose classification and detailed correctness evaluation. Each approach was designed to offer varying levels of integration, complexity, and specific advantages in their assessment methodologies.

1) *Approach 1: YOLOv11 Pose Classifier with MediaPipe Keypoint Evaluation (Hybrid Rule-Based)*: This approach adopts a distinct two-stage methodology, first employing a YOLOv11 model for high-level pose classification, followed by the integration of MediaPipe for granular keypoint extraction and a rule-based evaluation of pose correctness.

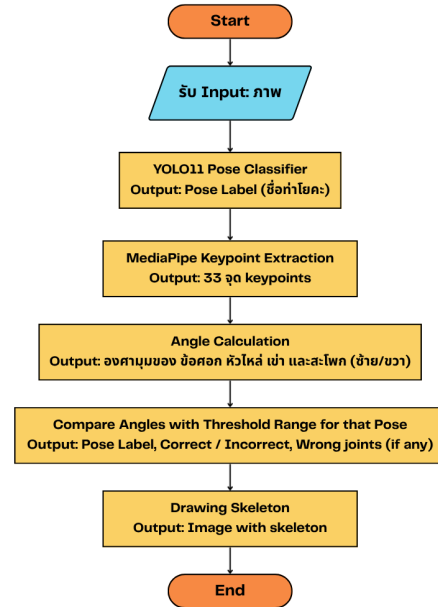


Figure 8. Flowchart illustrating the step-by-step process of Approach 1 in our proposed approach.

- 1) **Dataset Preparation and Model Training (YOLOv11 Classifier)**: Yoga pose images, sourced from Roboflow and categorized into classes such as *tree*, *warrior2*, *downdog*, *plank*, and *goddess*, were structured for training. A YOLOv11 object detection model (using ‘yolo11m.pt’ as its base) was trained on this dataset for 100 epochs. The sole objective of this trained model was to classify the overall type of yoga pose present within an image or frame.
- 2) **Human Pose Keypoint Extraction (MediaPipe)**: Subsequent to pose classification by YOLOv11, the MediaPipe Pose solution is actively invoked. It extracts 33

detailed body landmarks, provided as normalized coordinates, from the detected region of interest. Particular emphasis is placed on major joints crucial for yoga assessment (shoulders, elbows, hips, knees, ankles), with facial and foot landmarks being typically disregarded to simplify precise angle calculations.

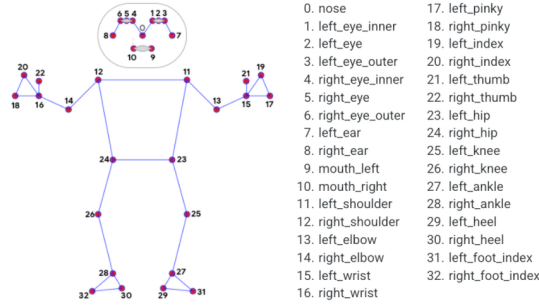


Figure 9. The 33 body landmarks detected by MediaPipe Pose, which form the basis for our geometric analysis. The list on the right maps each index number to its corresponding body part.

- 3) **Angle Calculation:** Based on the extracted MediaPipe landmarks, specific 2D joint angles for critical body segments (e.g., left/right elbow, shoulder, hip, knee) are robustly computed. Landmark coordinates, initially normalized, are converted to pixel coordinates relative to the detected person's bounding box to ensure highly accurate angular measurements via vector mathematics.
- 4) **Pose Side Detection and Variant Classification:** For inherently asymmetrical poses, such as Tree Pose or Warrior II, a sophisticated heuristic method is applied. This method primarily relies on comparing immediate joint angles (most notably knee angles) to dynamically determine if the left or right variant of the pose is being performed. This crucial step precisely categorizes the detected pose into specific variants like *tree_left* or *warrior2_right*, enabling significantly more nuanced and accurate correctness checks.
- 5) **Pose Correctness Evaluation (Rule-Based):** Pose correctness is evaluated using a rule-based system founded on angular thresholds. These thresholds were empirically defined by the 5th (P5) and 95th (P95) percentile range, derived from analyzing 'correct' pose samples in the Roboflow dataset (detailed in Table III). A joint is flagged as misaligned if its angle falls outside this data-driven range, triggering custom feedback. These values establish the system's **baseline parameters**, providing a robust, data-validated foundation for any future fine-tuning. The mean and standard deviation from these distributions provided an initial, data-driven baseline for the acceptable angular range of each joint.¹

¹The complete set of distribution plots is available for review in the following shared folder: https://drive.google.com/drive/folders/1NvqFkL9a80QJ00G3xp0Re-pgT5lTnXSI?usp=drive_link.

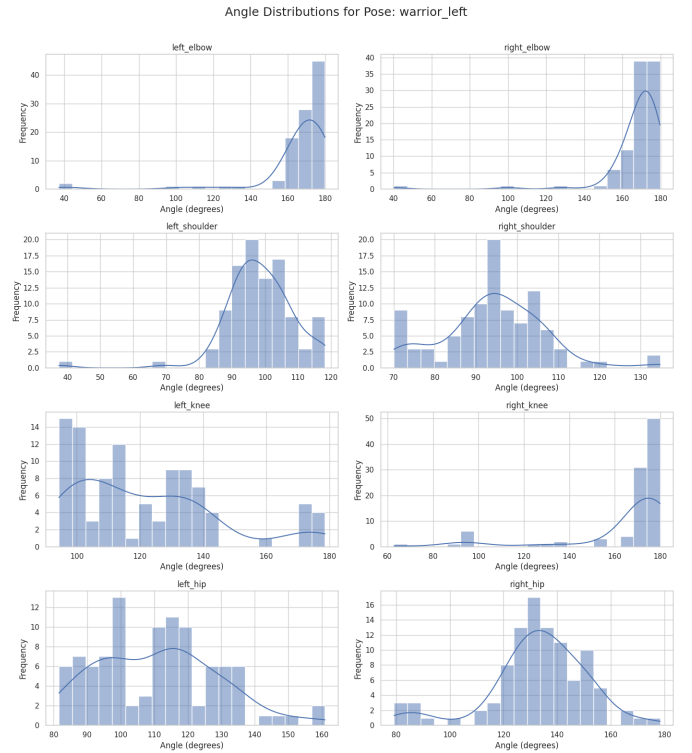


Figure 10. Example of the angle distribution for the left shoulder in the 'Warrior II (Right)' pose, generated from 'correct' samples in the Roboflow dataset.

Table III
EMPIRICALLY DERIVED ANGULAR THRESHOLDS (5TH AND 95TH PERCENTILES) FOR EACH POSE, BASED ON THE ROBOFLOW DATASET.

Pose Variant	Joint	Min Angle (P5, °)	Max Angle (P95, °)
Downward Dog	Left Elbow	147.83	178.47
	Left Hip	38.80	114.95
	Left Knee	133.46	179.45
	Left Shoulder	149.73	179.14
	Right Elbow	155.65	178.10
	Right Hip	42.14	107.66
	Right Knee	161.56	179.37
	Right Shoulder	102.61	179.19
Goddess	Left Elbow	11.78	69.79
	Left Hip	83.93	134.61
	Left Knee	86.24	140.76
	Left Shoulder	1.66	33.48
	Right Elbow	20.01	69.37
	Right Hip	86.84	156.46
	Right Knee	82.09	174.51
	Right Shoulder	1.99	34.95
Plank	Left Elbow	166.78	179.26
	Left Hip	147.73	179.24
	Left Knee	167.01	179.68
	Left Shoulder	51.93	78.51
	Right Elbow	153.10	179.32
	Right Hip	143.09	179.67
	Right Knee	156.50	179.20
	Right Shoulder	52.34	76.59
Tree (Left)	Left Elbow	12.80	58.41
	Left Hip	97.24	145.09
	Left Knee	21.64	51.99
	Left Shoulder	11.64	42.67
	Right Elbow	31.57	60.75
	Right Hip	172.69	179.84
	Right Knee	170.08	179.81
	Right Shoulder	3.56	52.16
Tree (Right)	Left Elbow	26.89	52.17
	Left Hip	138.43	179.82
	Left Knee	48.99	179.65
	Left Shoulder	13.27	50.35
	Right Elbow	23.91	49.01
	Right Hip	97.62	145.97
	Right Knee	18.49	50.36
	Right Shoulder	9.44	39.27
Warrior II (Left)	Left Elbow	131.90	179.49
	Left Hip	84.73	137.26
	Left Knee	97.32	172.85
	Left Shoulder	86.16	116.71
	Right Elbow	154.88	179.23
	Right Hip	88.49	154.24
	Right Knee	94.31	179.43
	Right Shoulder	72.59	110.36
Warrior II (Right)	Left Elbow	158.94	178.77
	Left Hip	110.73	153.58
	Left Knee	114.86	179.51
	Left Shoulder	79.07	109.89
	Right Elbow	157.01	179.57
	Right Hip	82.46	165.73
	Right Knee	97.55	170.41
	Right Shoulder	86.48	118.49

- 6) **Visualization and Output:** The extracted landmarks and their corresponding skeletal connections are visually overlaid onto the processed image or video frame. Correctly aligned joints and their connections are distinctly highlighted in green, while any misaligned ones are prominently marked in red. Ignored or missing joints are typically rendered in white. This rich visual feedback, synergistically combined with textual correction messages, is presented intuitively to the user. Annotated images or processed video frames are systematically saved for detailed review and analytical purposes.
- 7) **Batch Processing:** The system workflow is efficiently designed to support the processing of multiple static images from a specified directory or continuous, analysis

of video streams frame-by-frame, applying the described sequence of classification, keypoint extraction, angle calculation, comprehensive correctness evaluation, and dynamic visualization.

2) *Approach 2: YOLOv11 Person Detection + MediaPipe Keypoints + ML-based Hybrid Correctness:* This constitutes our most advanced and robust approach, integrating precise human detection from YOLOv11 with MediaPipe’s highly accurate keypoint extraction, followed by a sophisticated multi-model machine learning system. This system performs both initial pose classification and a meticulously refined hybrid correctness assessment.

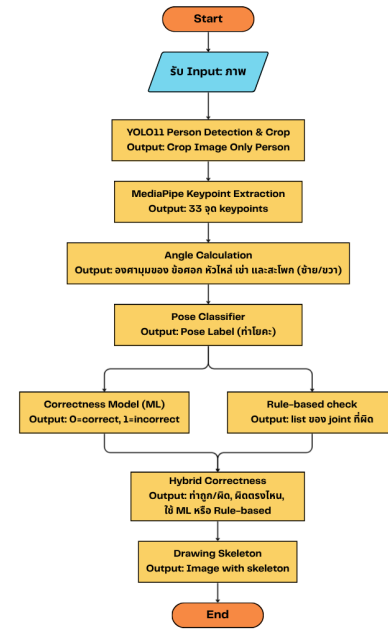


Figure 11. Flowchart illustrating the step-by-step process of Approach 2 in our proposed approach.

- 1) **Human Detection (YOLOv11):** An off-the-shelf YOLOv11 model, pre-trained on the comprehensive COCO dataset, is employed for the initial and crucial task of detecting human figures (‘person’ class) within the input image or video frame. In scenarios involving multiple detected individuals, the largest bounding box (heuristically assumed to enclose the primary subject of interest) is strategically selected. The content within this selected bounding box is then cropped for subsequent detailed processing, optimizing computational focus.
- 2) **Pose Keypoint Extraction (MediaPipe):** For the cropped human figure isolated in the previous step, MediaPipe Pose (configured for ‘static-image-mode=True’ for single-frame processing) is utilized to extract 33 detailed 2D body landmarks. Consistent with Approach 2, only major joints critically relevant to yoga evaluation (e.g., shoulders, elbows, hips, knees, ankles) are further considered, while less pertinent landmarks are filtered

out to maintain focus and efficiency.

- 3) **Angle Calculation:** From the 2D coordinates of the extracted keypoints, a set of 8 essential joint angles is precisely computed: encompassing the left/right elbow, shoulder, hip, and knee angles. These calculations are performed using robust vector mathematics, effectively translating pixel-level coordinates into quantifiable angular measurements, crucial for objective assessment.²

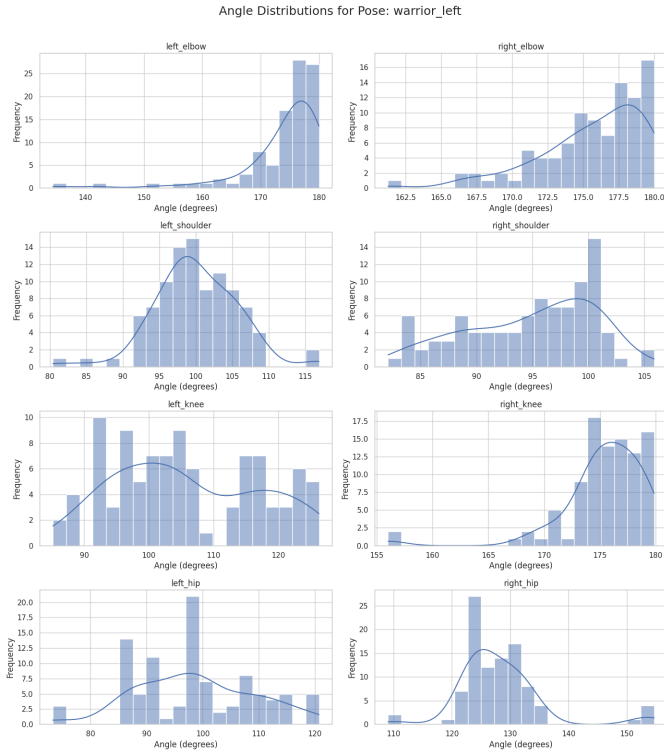


Figure 12. Example of the angle distribution for the left shoulder in the 'Warrior II (Right)' pose, generated from 'correct' samples in the Kaggle dataset.

²The complete set of distribution plots is available for review in the following shared folder: https://drive.google.com/drive/folders/1NvqFkL9a80QJ00G3xp0Re-pgT5ITnXSI?usp=drive_link.

Table IV
EMPIRICALLY DERIVED ANGULAR THRESHOLDS (5TH AND 95TH PERCENTILES) FROM THE KAGGLE DATASET.

Pose Variant	Joint	Min Angle (P5, °)	Max Angle (P95, °)
Downward Dog	Left Elbow	155.63	178.66
	Left Hip	64.18	90.41
	Left Knee	169.67	179.57
	Left Shoulder	151.72	177.73
	Right Elbow	154.80	177.84
	Right Hip	62.96	91.75
	Right Knee	170.63	179.68
	Right Shoulder	152.81	179.49
Goddess	Left Elbow	15.13	111.25
	Left Hip	87.17	123.91
	Left Knee	84.34	129.55
	Left Shoulder	13.69	108.90
	Right Elbow	7.82	128.76
	Right Hip	83.56	120.92
	Right Knee	81.16	126.48
	Right Shoulder	9.15	118.13
Plank	Left Elbow	79.36	178.73
	Left Hip	149.33	178.95
	Left Knee	166.60	179.58
	Left Shoulder	60.65	101.31
	Right Elbow	74.15	178.95
	Right Hip	155.98	178.67
	Right Knee	168.56	179.67
	Right Shoulder	66.20	98.90
Tree (Left)	Left Elbow	145.03	170.13
	Left Hip	113.55	173.93
	Left Knee	22.06	177.05
	Left Shoulder	172.22	179.93
	Right Elbow	140.57	171.96
	Right Hip	110.72	179.54
	Right Knee	28.98	179.82
	Right Shoulder	166.37	179.67
Tree (Right)	Left Elbow	90.46	174.03
	Left Hip	170.62	179.75
	Left Knee	27.04	179.85
	Left Shoulder	161.31	178.64
	Right Elbow	90.12	176.04
	Right Hip	104.98	127.03
	Right Knee	21.19	61.40
	Right Shoulder	152.53	179.67
Warrior II (Left)	Left Elbow	161.13	179.30
	Left Hip	85.46	115.17
	Left Knee	88.92	124.11
	Left Shoulder	92.40	108.09
	Right Elbow	168.02	179.58
	Right Hip	122.17	139.11
	Right Knee	169.95	179.46
	Right Shoulder	84.37	101.20
Warrior II (Right)	Left Elbow	91.07	179.24
	Left Hip	102.36	141.48
	Left Knee	109.64	179.83
	Left Shoulder	78.86	106.24
	Right Elbow	164.05	178.98
	Right Hip	80.39	127.95
	Right Knee	95.79	176.65
	Right Shoulder	87.29	111.59

- 4) **Pose Side Detection and Variant Classification:** For inherently asymmetrical yoga poses (e.g., Tree Pose, Warrior II), a heuristic rule is applied. By comparing specific calculated knee angles (e.g., the degree of flexion), the system accurately determines the 'active' side (left or right) and assigns a precise pose variant label (e.g., *tree-left*, *warrior2-right*). For symmetrical poses such as Downward Dog, Plank, and Goddess Pose, the base pose name is used directly without side differentiation.

5) Pose Classification (ML-based):

- a) **Dataset Preparation:** The 8 computed joint angles for all correctly performed yoga poses within our training dataset (including specific variations

like *tree-left*) are systematically extracted. These numerical features are then meticulously stored in a CSV file, paired with their corresponding ground-truth pose labels, forming the supervised learning dataset.

- b) **Model Training:** A 'RandomForestClassifier' is trained using these angle features ('left-knee', 'right-knee', 'left-elbow', 'right-elbow', 'left-hip', 'right-hip', 'left-shoulder', 'right-shoulder') to predict the specific yoga pose variant. The model undergoes rigorous training on the prepared dataset, and its performance is evaluated using standard classification metrics such as precision, recall, and F1-score. The optimized trained model ('pose-classifier.pkl') is then serialized using 'joblib' for efficient and rapid inference.
 - c) **Inference:** For any new, unseen image input, the 8 computed joint angles (derived from steps 2 and 3) are fed as input features into the pre-trained 'RandomForestClassifier'. This process yields a predicted pose label, such as *downdog* or *tree-left*.
- 6) **Hybrid Pose Correctness Evaluation:** This is the cornerstone of our advanced system, employing a dual-layer approach for discerning pose correctness.
- a) **Correctness Model Training (Gradient Boosting):** To assess the granular correctness of a predicted pose, a dedicated 'GradientBoostingClassifier' is trained individually for each distinct yoga pose variant. For instance, for the *tree-left* pose, this classifier learns to robustly distinguish between correct (labeled 0) and incorrect (labeled 1) executions. The training data for these specialized models is comprehensively generated from:
 - **Real Correct Samples:** Angles extracted from naturally occurring correct pose images.
 - **Real Incorrect Samples:** Angles extracted from user-generated or curated incorrect pose images.
 - **Synthetically Augmented Incorrect Samples:** To address potential data scarcity and enhance model robustness, synthetic incorrect samples are generated by introducing controlled Gaussian noise to the real incorrect samples. This augmentation is carefully constrained within physically plausible angular limits (0-180 degrees) to ensure realism and prevent the generation of unfeasible poses.

A 'sample weight' mechanism is applied during training to assign a higher importance to real incorrect examples, thereby improving the model's sensitivity to critical errors. Each individually trained correctness model is subsequently saved via joblib.
 - b) **Rule-based Verification:** Concurrently with the ML-based assessment, a robust rule-based system is employed. This system leverages predefined angle thresholds (minimum and maximum angular

ranges) for each critical joint, meticulously customized for every specific pose variant. If a joint's calculated angle deviates outside its established acceptable range, it is unequivocally flagged as misaligned, providing a foundation for interpretable feedback.

- c) **Hybrid Decision Logic:** The system intelligently combines the probabilistic decision from the ML-based 'GradientBoostingClassifier' with the deterministic insights from the rule-based verification:
 - **ML Predicts Correct, Rules Confirm:** If the ML model predicts 'correct' AND the rule-based check finds absolutely no misaligned joints, the pose is confidently deemed 'correct'.
 - **ML Predicts Incorrect, Rules Indicate Errors:** If the ML model predicts 'incorrect' AND the rule-based check independently identifies specific joint errors, the pose is classified as 'incorrect', and precise feedback regarding those joints is provided.
 - **ML Predicts Incorrect, Rules Find No Errors (Rule Override):** In instances where the ML model predicts 'incorrect' but the rule-based check reveals no angular deviations (potential ML false positive), the rule-based check is prioritized, and the pose is re-classified as 'correct'. This mechanism enhances user trust and accounts for the strictness of rule-based validity.
 - **ML Predicts Correct, Rules Indicate Errors (Threshold Secure):** Conversely, if the ML model predicts 'correct' yet the rule-based check identifies clear angular errors (potential ML false negative), the strict rule-based check is prioritized. The pose is then classified as 'incorrect', and feedback for the misaligned joints is furnished. This conservative approach safeguards against potentially risky false positives where user safety or correct form is paramount.
- d) **Visualization and Feedback:** In the final output presentation, any joints identified as misaligned (through this hybrid decision process) are prominently highlighted in red on the visualized skeletal overlay, while correctly aligned joints are distinctly shown in green. The system delivers comprehensive and user-friendly feedback, which includes:
 - The predicted yoga pose variant (e.g., "Tree Left").
 - An overall correctness status (Boolean: True/False).
 - A precise list of all identified misaligned joints.
 - The decision source field, explicitly indicating whether the outcome was primarily driven by ML, rule-based logic, or a hybrid combination.
 - The computed joint angles and the raw extracted keypoints for detailed review.

Collectively, these two distinct approaches provide a versatile suite of options for effective pose detection and evaluation. Approach 1, which combines a YOLOv11 pose classifier with a rule-based MediaPipe evaluation, offers a robust two-stage pipeline where high-level pose identification precedes detailed geometric verification. Approach 2, our most advanced hybrid system, completely decouples detection, classification, and correctness assessment by leveraging a specialized YOLOv11 person detector and a multi-layered machine learning framework. This modularity delivers superior accuracy and granular, interpretable feedback, making it ideal for detailed posture analysis and corrective guidance. This flexibility renders our overall system highly adaptable for real-time applications for future work. The current implementation fully supports inference on static images as well as robust live video analysis, delivering accurate pose classification and precise correctness assessment. Furthermore, the modular design inherently supports scalability for the inclusion of additional pose classes or the potential extension to multi-person detection scenarios in future work.

D. Performance Evaluation Metrics

The efficacy of the proposed system and its constituent approaches will be rigorously evaluated using a set of standard machine learning and computer vision metrics. These include:

- **Precision, Recall, and F1-Score:** To assess the accuracy of both pose classification and correctness assessment.
- **Mean Average Precision (mAP):** For evaluating the performance of the object detection components.
- **Frames Per Second (FPS):** To measure the real-time inference speed and computational efficiency of each approach.

These metrics will allow for a comprehensive and quantitative comparison between the different implemented methodologies.

IV. RESULTS

This section presents the experimental results for each of the two implemented approaches, evaluating their performance in yoga pose classification and correctness assessment.

A. Approach 1: YOLOv11 Pose Classifier with MediaPipe (Rule-Based Evaluation)

In this approach, the system first classifies the yoga pose using a fine-tuned YOLOv11 model and subsequently evaluates its correctness using a rule-based system powered by MediaPipe keypoints. Figures 13 to 15 illustrate the system's ability to differentiate between correct and incorrect executions, highlighting misaligned joints and connections in red based on predefined angle thresholds.

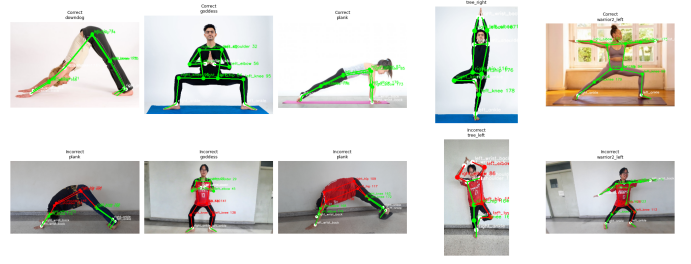


Figure 13. Visual feedback from the rule-based YOLOv11 + MediaPipe approach.

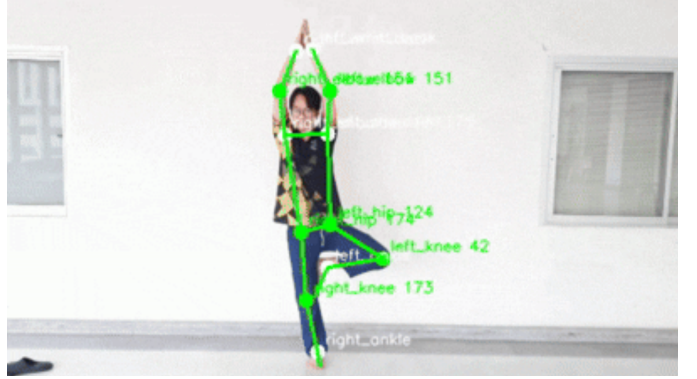


Figure 14. Visual (VDO Snapshot) feedback from the rule-based YOLOv11 + MediaPipe approach.

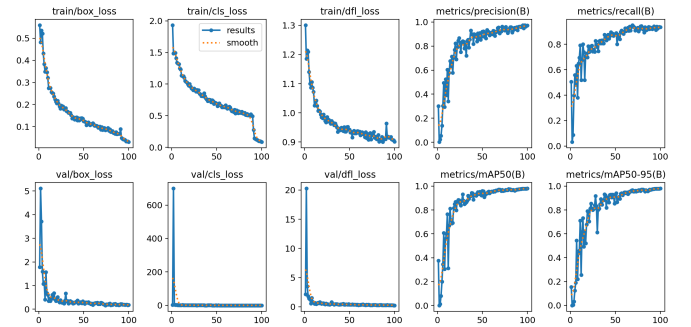


Figure 15. Visual feedback from the rule-based YOLOv11 + MediaPipe approach.

Analysis of Training and Validation Metrics. Figure 15 displays the model's training dynamics over 100 epochs, depicting a healthy and successful learning process.

Key observations include:

- **Effective Convergence and Generalization:** All loss functions for both training and validation decreased rapidly and stabilized at low values. The close alignment between the curves confirms excellent generalization with no significant signs of overfitting.
- **High Performance Metrics:** Key performance indicators, including precision, recall, and Mean Average Precision (mAP50 and mAP50-95), all show a strong convergence towards near-optimal values, validating the

model’s high accuracy in both localization and classification.

In summary, the training graphs confirm a stable and effective process, resulting in a robust and highly accurate model suitable for deployment.

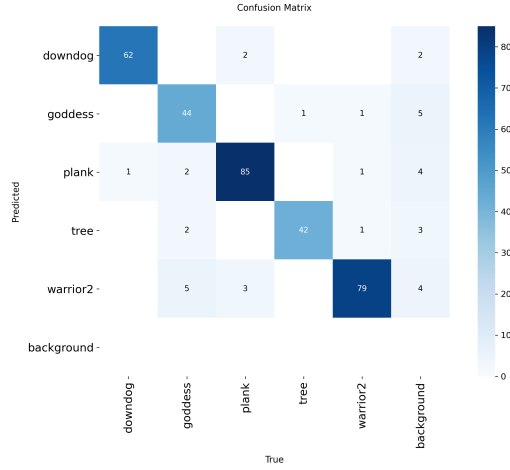


Figure 16. Visual feedback from the rule-based YOLOv11 + MediaPipe approach.

Analysis of the Confusion Matrix. Figure 16 presents the confusion matrix for our pose classification model. The strong diagonal values confirm the model’s high efficacy in classifying most primary yoga poses, particularly those with distinct silhouettes like `plank` and `downdog`. However, two key limitations are apparent. First, a notable confusion exists between the `goddess` and `warrior2` poses, likely due to similar limb angles. Second, and more critically, the model completely fails to identify the `background` class, consistently misclassifying non-pose images as valid postures. This indicates the model is over-optimized to assign a known pose label. Enhancing its real-world applicability will require enriching the training dataset with more diverse negative samples to improve its recognition of non-pose scenarios.

Table V
PERFORMANCE OF THE RULE-BASED EVALUATION (APPROACH 1).

Pose	Precision (%)	Recall (%)	mAP50
Overall	97.2	93.6	98.1
Downward Dog	98.0	98.4	97.9
Tree	97.7	98.2	99.3
Goddess	99.5	83.0	97.0
Plank	95.2	93.5	97.6
Warrior2	95.6	95.1	98.8

Discussion. As shown in Table V, the rule-based approach demonstrates impressive performance in pose classification, achieving a remarkable overall mAP50 of 98.1

The success of this two-stage process—classifying the pose first and then applying geometric rules—confirms its viability

for specific joint-level error detection. However, despite the high classification accuracy, its fundamental limitation lies in the rigidity of the hand-crafted angle thresholds used for correctness evaluation. These rules may not adequately generalize across diverse body types or subtle, yet acceptable, variations in posture, potentially leading to inaccuracies in the final correctness feedback.

B. Approach 2: YOLOv11 Person Detection with ML-based Hybrid Correctness

This advanced approach decouples human detection from pose evaluation, employing a sophisticated ML-based pipeline for both pose classification and correctness assessment. Figures 17 showcases the system’s enhanced output, where the model accurately identifies misaligned joints (in red) based on the hybrid decision of ML models and rule-based verification. The ‘decision source’ field, also visible in the output, provides transparency into the assessment process.

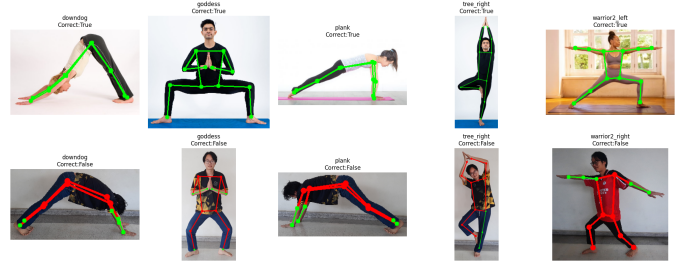


Figure 17. Visual feedback from the ML-based hybrid approach, highlighting specific joint errors and decision source.



Figure 18. Visual (VDO Snapshot) feedback from the ML-based hybrid approach, highlighting specific joint errors and decision source.

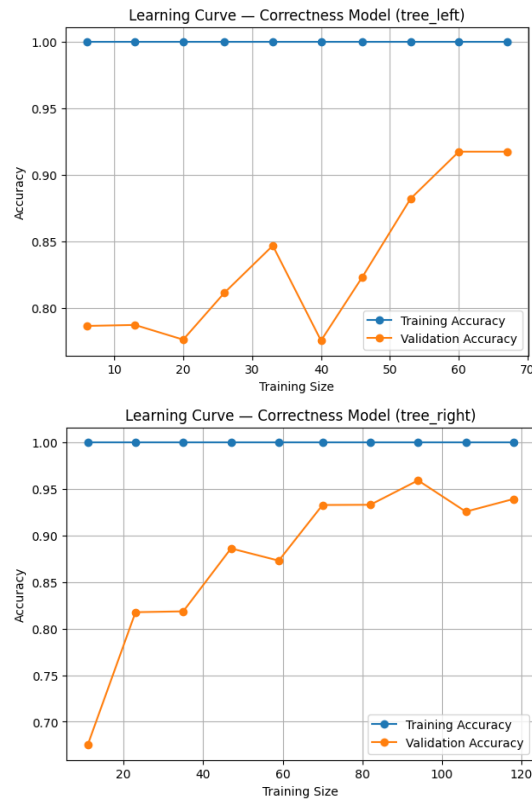


Figure 19. Learning Curves Tree Pose

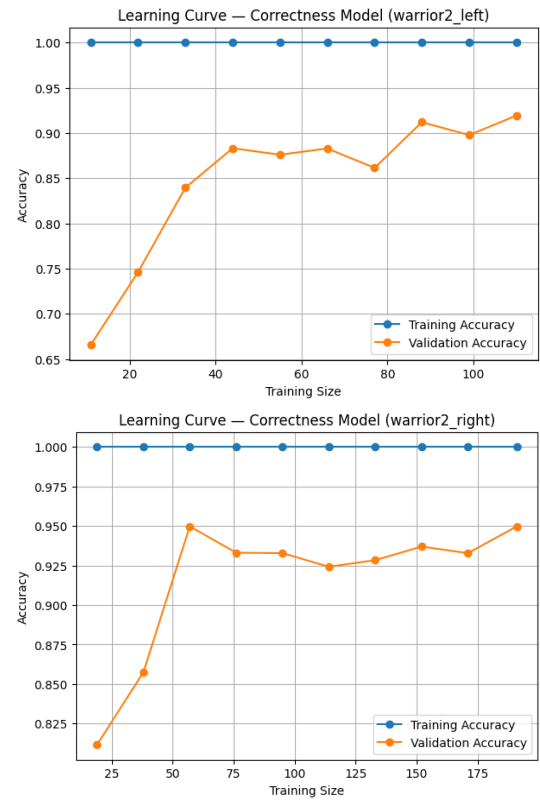


Figure 20. Learning Curves Warrior2 Pose

Analysis of Learning Curves for Tree Pose Correctness Models. Figure 19 displays the learning curves for the *tree_right* and *tree_left* correctness models. These curves offer insights into the models' learning behavior as a function of training set size.

The graphs reveal several key characteristics:

- **High Model Capacity:** For both variants, the training accuracy consistently remains at 1.0, indicating the models have sufficient capacity to fit the training data perfectly.
- **Effective Generalization with More Data:** The validation accuracy for both models shows a clear upward trend as more training data is introduced. For *tree_right*, it climbs to a peak of 0.96, while for *tree_left*, despite some volatility, it reaches over 0.91. This demonstrates that more data directly improves generalization.
- **High Initial Variance:** The initial wide gap between the perfect training accuracy and the lower validation accuracy is a classic sign of high variance. However, this gap narrows as the dataset size increases, confirming that more data is an effective strategy to mitigate overfitting for these models.

In conclusion, the learning curves confirm that the Gradient Boosting models are well-suited for this task. Their performance improves substantially with more data, validating that our data-driven approach to correctness assessment is both viable and effective.

Analysis of Learning Curves for Warrior II Correctness Models.

Figure 20 shows that the *warrior2_right* and *warrior2_left* models learn effectively as the training set size increases.

- **Effective Learning:** Perfect training accuracy (1.0) indicates high model capacity, while validation accuracy rises to 0.95 and 0.91 respectively, showing strong generalization with more data.
- **Variance Reduction:** The narrowing gap between the training and validation curves confirms that a larger dataset successfully mitigates initial overfitting.

In conclusion, the learning curves validate our data-driven approach, demonstrating that model performance on the complex Warrior II pose scales effectively with data volume.

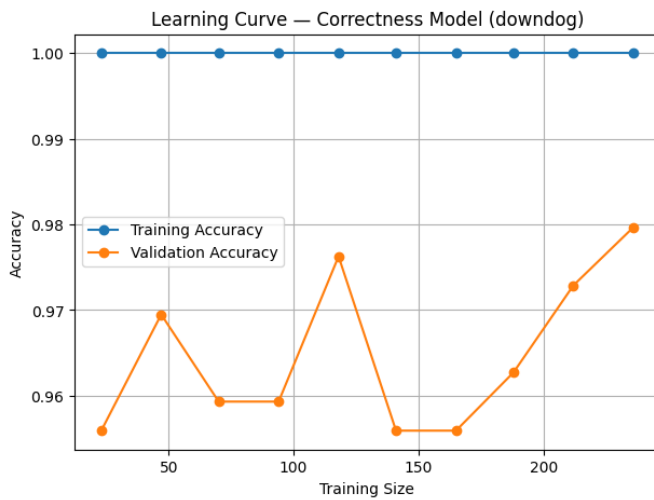


Figure 21. Learning Curves Downward Dog Pose

Analysis of the Learning Curve for the Downward Dog Correctness Model. Figure 21 presents the learning curve for the correctness model of the *Downward Dog* pose, revealing a highly effective model with some unique learning characteristics.

The key observations from this curve are as follows:

- **Minimal Variance and Excellent Generalization:** The validation accuracy remains exceptionally high (fluctuating between 0.955-0.98) and closely tracks the perfect training accuracy. This small gap indicates very low variance and strong generalization, suggesting the features defining the pose are highly distinct and easily learned.
- **High but Volatile Validation Performance:** Despite its high overall performance, the validation curve exhibits noticeable volatility, with sharp peaks and troughs. This could be attributed to the model's sensitivity to specific 'hard' or ambiguous examples within the cross-validation folds.
- **Consistently High Accuracy:** Even with the volatility, the validation accuracy consistently remains above an excellent threshold of 95.5%. The curve's final upward trajectory suggests that performance continues to improve as the dataset grows.

In summary, the learning curve indicates a highly effective and well-generalized classifier for the *Downward Dog* pose. While the validation volatility is a point of academic interest, the consistently high accuracy confirms the model's strong reliability for this assessment task.

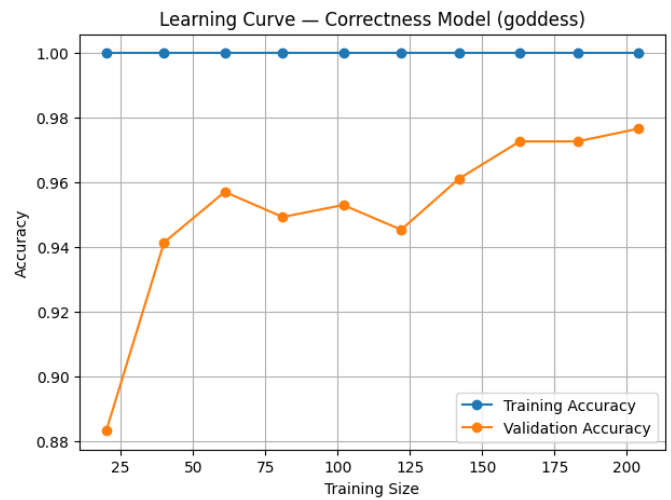


Figure 22. Learning Curves Goddess Pose

Analysis of the Learning Curve for the Goddess Pose Correctness Model. Figure 22 illustrates the learning curve for the correctness assessment model dedicated to the Goddess pose. This graph shows the model's accuracy on both training and validation sets as a function of the training data size.

The learning curve for the Goddess pose model reveals a classic and healthy training progression:

- **Perfect Training Accuracy and Model Capacity:** The **Training Accuracy** consistently maintains a perfect score of 1.0. This demonstrates that the model has more than enough capacity to fully learn and memorize the features present in the training data, regardless of the sample size.
- **Clear Generalization Improvement:** The **Validation Accuracy** shows a strong and positive correlation with the amount of training data. It begins at approximately 0.88 and steadily climbs to a final accuracy of nearly 0.98. This upward trend clearly indicates that as the model is exposed to more examples, its ability to generalize to unseen data improves significantly.
- **Healthy Convergence and Mitigated Variance:** A key feature of this curve is the gradual convergence between the training and validation lines. The initial gap between the two curves, which signifies high variance (overfitting to the small training set), progressively narrows as the training size increases. This convergence is a strong indicator of a well-behaving model that benefits directly from a larger dataset to reduce variance and enhance its generalization capabilities.
- **Stable Learning Trajectory:** Despite minor fluctuations, the validation curve follows a generally stable and increasing path. This suggests that the learning process is robust and that the features defining a correct or incorrect Goddess pose are well-defined and learnable by the model.

In conclusion, the learning curve for the Goddess pose correctness model exemplifies a successful training process.

It confirms that the model is well-suited for the task and that its performance is directly scalable with the size of the training dataset, leading to a highly accurate and reliable correctness assessment.

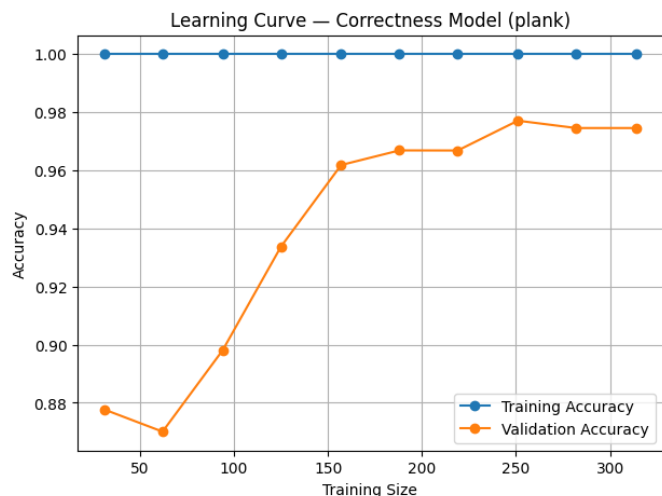


Figure 23. Learning Curves Warrior2 Pose

Analysis of the Plank Pose Learning Curve.

Figure 23 displays a classic and effective learning progression for the Plank pose model, demonstrating that its performance scales directly with data volume.

- **Effective Learning and Generalization:** The model exhibits high capacity with perfect training accuracy (1.0). As the training set grows, validation accuracy climbs sharply to nearly 0.98, which significantly narrows the initial high-variance (overfitting) gap.
- **Performance Plateau:** The validation curve begins to plateau near its peak, suggesting the model is approaching its optimal performance and that further gains may require strategies beyond simply increasing data volume.

In summary, the learning curve confirms a successful, data-hungry learning process. Increasing the dataset was the key factor in resolving initial overfitting and enabling the model to achieve excellent generalization on the Plank pose.

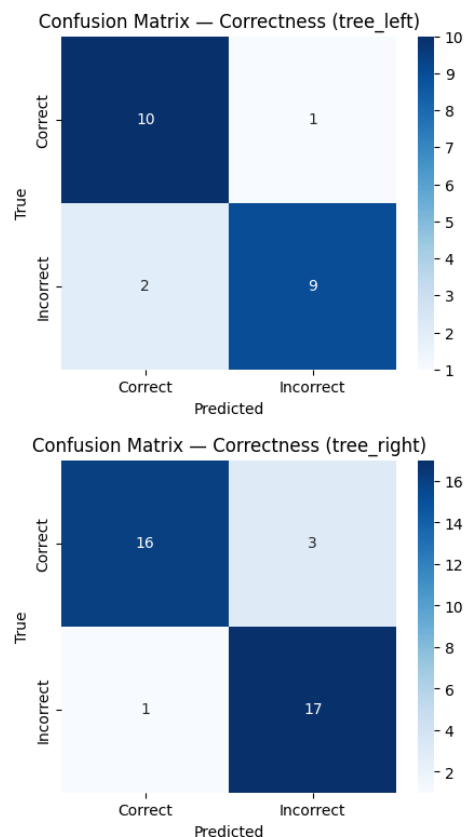


Figure 24. Confusion Matrix Tree Pose

Analysis of Confusion Matrices for Tree Pose Correctness Models. Figure 24 provides the confusion matrices for the correctness assessment models of both the `tree_right` and `tree_left` variants. These matrices detail the models' performance in distinguishing between 'Correct' and 'Incorrect' executions on the test set.

a) Tree Right Variant

The confusion matrix for the **tree_right** model demonstrates strong performance. Out of 19 'Correct' instances, the model accurately identified 16 (True Positives) and misclassified only 3 as 'Incorrect' (False Negatives). For the 'Incorrect' class, it correctly identified 17 out of 18 instances (True Negatives) and made only a single error by misclassifying an 'Incorrect' pose as 'Correct' (False Positive). The low number of false positives is particularly important, as it indicates the model is reliable in not approving incorrect postures.

b) Tree Left Variant

Similarly, the correctness model for the **tree_left** variant also shows effective classification capabilities. It correctly identified 10 out of 11 'Correct' instances, with only 1 false negative. For the 'Incorrect' class, it correctly identified 9 out of 11 instances, with 2 false positives. While slightly less precise than its 'right' counterpart due to the two false positives, the model still maintains a high level of accuracy.

c) Overall Assessment

Both models exhibit a high degree of accuracy and a balanced ability to identify both correct and incorrect poses. The

total number of misclassifications is small for both variants (4 for 'tree_right' and 3 for 'tree_left'), confirming the robustness of our ML-based correctness assessment approach for the asymmetrical Tree Pose. The results validate that the models have learned to effectively discern the nuanced geometric differences between a properly executed pose and its common errors.

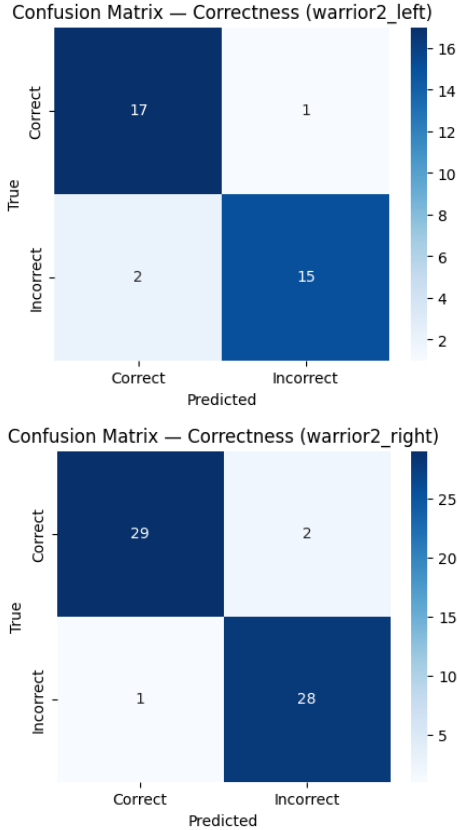


Figure 25. Confusion Matrix Warrior2 Pose

Analysis of Confusion Matrices for Warrior II Correctness Models. Figure 25 presents the confusion matrices for the correctness assessment models of the warrior2_right and warrior2_left pose variants. These matrices provide a detailed breakdown of the models' performance in classifying 'Correct' versus 'Incorrect' executions on the test set.

d) Warrior II Right Variant

The model for the **warrior2_right** variant demonstrates exceptional accuracy. It correctly identified 29 out of 31 'Correct' instances (True Positives), with only 2 false negatives. For the 'Incorrect' class, the performance was even more robust, correctly identifying 28 out of 29 instances (True Negatives) while making only a single error by misclassifying an 'Incorrect' pose as 'Correct' (False Positive). With only 3 misclassifications in total across 60 samples, this model proves to be highly reliable.

e) Warrior II Left Variant

The **warrior2_left** model also exhibits strong, albeit slightly less precise, performance. It correctly identified 17 out of

18 'Correct' instances, with only 1 false negative. In the 'Incorrect' class, it correctly identified 15 out of 17 instances but misclassified 2 incorrect poses as correct (False Positives). These two false positive errors are the primary reason for the slightly lower precision observed for this variant in the overall performance table.

f) Overall Assessment

Both correctness models for the Warrior II pose show a high level of performance and a strong ability to distinguish between correct and incorrect forms. The 'warrior2_right' model is particularly effective, achieving near-perfect classification. The slightly higher number of false positives in the 'warrior2_left' model suggests that some incorrect variations of this pose may be geometrically closer to the correct form, making them harder for the model to distinguish. Nevertheless, the overall results confirm that the ML-based correctness assessment is a powerful and effective tool for this complex, asymmetrical pose.

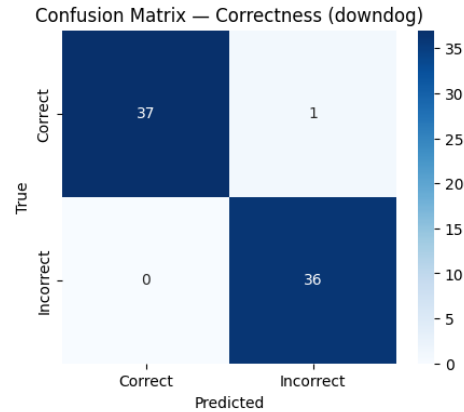


Figure 26. Learning Curves Downward Dog Pose

Analysis of the Confusion Matrix for the Downward Dog Correctness Model. Figure 26 presents the confusion matrix for the correctness assessment model of the Downward Dog pose. The results demonstrate an exceptionally high level of accuracy and reliability for this specific task.

The key takeaways from the matrix are as follows:

- **Near-Perfect Performance:** The model exhibits near-perfect classification capabilities across a total of 74 test samples. It correctly identified 37 out of 38 'Correct' instances and all 36 'Incorrect' instances without error.
- **Zero False Positives:** A critical and outstanding result is the complete absence of false positives (the cell representing 'True: Incorrect, Predicted: Correct' is 0). This means the model never misclassified an 'Incorrect' pose as 'Correct'. This is a highly desirable characteristic for a fitness application, as it ensures that users are not given misleading positive feedback on improper form, thereby prioritizing safety and effective learning.
- **Minimal False Negatives:** There was only a single instance where the model misclassified a 'Correct' pose as 'Incorrect' (a false negative). While ideally zero, this

single error out of 74 samples is exceptionally low and points to a very high recall for the 'Correct' class.

In summary, the confusion matrix for the Downward Dog correctness model validates its outstanding performance. The model is not only highly accurate but also demonstrates a crucial conservative bias, making it extremely reliable for real-world application by preventing the approval of incorrect postures. This result is consistent with the high and stable validation accuracy observed in its learning curve.

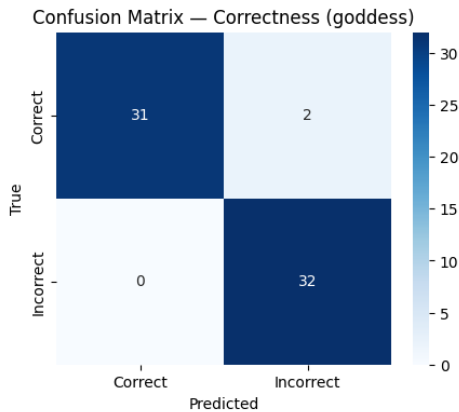


Figure 27. Confusion Matrix Goddess Pose

Analysis of the Confusion Matrix for the Goddess Pose Correctness Model. Figure 27 presents the confusion matrix for the correctness assessment model of the Goddess pose. This matrix clearly demonstrates the model's exceptional performance and reliability in distinguishing between correct and incorrect executions of this pose.

Key findings from the matrix include:

- **Outstanding Overall Accuracy:** With only 2 misclassifications out of 65 total test samples, the model achieves an exceptionally high accuracy. It correctly identified 31 out of 33 'Correct' instances and achieved a perfect score on the 'Incorrect' class, correctly identifying all 32 instances.
- **Perfect Recall for the 'Incorrect' Class (Zero False Positives):** A standout result is the model's flawless performance in identifying incorrect poses. It made zero false positive errors, meaning it never misclassified an 'Incorrect' pose as 'Correct'. This perfect recall for the 'Incorrect' class underscores the model's reliability and safety, ensuring users are never given false assurance on improper form.
- **Very Low False Negatives:** The model made only two errors in total, both of which were false negatives (misclassifying a 'Correct' pose as 'Incorrect'). This indicates a slight conservative bias, which is preferable in a fitness application, as it prioritizes caution over potentially approving a borderline or incorrect posture.

In conclusion, the confusion matrix for the Goddess pose correctness model confirms its superb efficacy. The model is not only highly accurate but also highly reliable, particularly

in its perfect identification of incorrect postures. This result aligns with the high and stable validation accuracy observed in its learning curve, validating the model's suitability for real-world deployment.

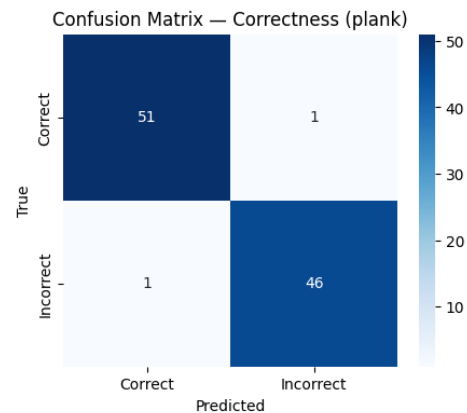


Figure 28. Confusion Matrix Plank Pose

Analysis of the Confusion Matrix for the Plank Pose Correctness Model. Figure 28 presents the confusion matrix for the correctness assessment model of the Plank pose. The results highlight the model's exceptional accuracy and balanced performance in distinguishing between correct and incorrect executions.

Key insights drawn from the matrix are:

- **Extremely High Accuracy and Balanced Performance:** The model demonstrates remarkable accuracy, with only two misclassifications across a total of 99 test samples. It correctly identified 51 out of 52 'Correct' instances and 46 out of 47 'Incorrect' instances. This near-perfect balance indicates that the model is equally proficient at identifying both proper and improper form.
- **Minimal Errors:** The model made only one false positive error (misclassifying an 'Incorrect' pose as 'Correct') and one false negative error (misclassifying a 'Correct' pose as 'Incorrect'). Such a low and symmetrical error rate is indicative of a well-calibrated and robust classifier that has effectively learned the distinguishing features of the Plank pose.
- **High Reliability for Practical Use:** The extremely low number of an incorrect pose being approved (one false positive) ensures a high degree of safety and reliability for practical application. Users can trust the system's feedback, as it is highly unlikely to provide false assurance on an incorrect posture.

In conclusion, the confusion matrix for the Plank pose correctness model confirms its outstanding efficacy. The model is both highly accurate and well-balanced, making it a reliable tool for providing automated feedback. This strong performance is consistent with the high validation accuracy achieved in its learning curve, validating the model's robustness for this specific task.

Table VI
CORRECTNESS MODEL FOR TREE_LEFT

Pose	Precision	Recall	F1 Score
0	0.83	0.91	0.87
1	0.90	0.82	0.86
Accuracy			0.86
Macro accuracy	0.87	0.86	0.86
Weighted accuracy	0.87	0.86	0.86

Table VII
CORRECTNESS MODEL FOR TREE_RIGHT

Pose	Precision	Recall	F1 Score
0	0.94	0.84	0.89
1	0.85	0.94	0.89
Accuracy			0.89
Macro accuracy	0.90	0.89	0.89
Weighted accuracy	0.90	0.89	0.89

Quantitative Analysis of Tree Pose Correctness Models. Tables VI and VII present the detailed classification performance for the correctness models of the *tree_left* and *tree_right* variants, respectively. These metrics provide a quantitative measure of the models’ ability to distinguish between correct (class 0) and incorrect (class 1) executions.

g) *Tree Left Variant*

The model for the *tree_left* variant achieves a high overall accuracy of 86%, as shown in Table VI. The model demonstrates a good balance between precision and recall. For the ‘Correct’ class (0), it achieves a high recall of 0.91, indicating it is very effective at identifying correctly performed poses. However, its precision for this class is slightly lower at 0.83, suggesting it occasionally misclassifies incorrect poses as correct. Conversely, for the ‘Incorrect’ class (1), it has a high precision of 0.90 but a lower recall of 0.82.

h) *Tree Right Variant*

The *tree_right* model shows a slightly stronger performance with an overall accuracy of 89%, as detailed in Table VII. This model also exhibits a well-balanced performance. Notably, it has a high recall of 0.94 for the ‘Incorrect’ class (1), signifying its excellent capability to detect improper postures, even though this comes at the cost of a slightly lower precision (0.85). The performance for the ‘Correct’ class (0) is also strong, with a high precision of 0.94.

i) *Overall Assessment*

Both models demonstrate robust performance with weighted F1-scores of 0.86 for *tree_left* and 0.89 for *tree_right*. The results indicate that the data-driven approach is highly effective for the complex task of correctness assessment for the asymmetrical Tree Pose. A notable trade-off between precision and recall can be observed in both models; for instance, the *tree_left* model is better at not missing correct poses (high recall for class 0), while the *tree_right* model is more adept at identifying incorrect poses (high recall for class 1). This nuanced performance highlights the models’ ability to learn

the subtle geometric differences characteristic of correct versus incorrect executions.

Table VIII
CORRECTNESS MODEL FOR WARRIOR2_LEFT

Pose	Precision	Recall	F1 Score
0	0.89	0.94	0.92
1	0.94	0.88	0.91
Accuracy			0.91
Macro accuracy	0.92	0.91	0.91
Weighted accuracy	0.92	0.91	0.91

Table IX
CORRECTNESS MODEL FOR WARRIOR2_RIGHT

Pose	Precision	Recall	F1 Score
0	0.97	0.94	0.95
1	0.93	0.87	0.95
Accuracy			0.95
Macro accuracy	0.95	0.95	0.95
Weighted accuracy	0.95	0.95	0.95

Quantitative Analysis of Warrior II Pose Correctness Models. Tables VIII and IX detail the classification performance for the correctness models of the *warrior2_left* and *warrior2_right* variants, respectively. These metrics quantify the models’ ability to differentiate between correct (class 0) and incorrect (class 1) executions.

j) *Warrior II Left Variant*

The model for the *warrior2_left* variant achieves a high overall accuracy of 91%, as shown in Table VIII. It demonstrates a strong and well-balanced performance, with a weighted F1-score of 0.91. For the ‘Correct’ class (0), the model has a very high recall of 0.94, indicating its effectiveness in successfully identifying most correctly performed poses. For the ‘Incorrect’ class (1), it achieves a high precision of 0.94, signifying that when the model predicts a pose as incorrect, it is highly likely to be right.

k) *Warrior II Right Variant*

The *warrior2_right* model exhibits even stronger performance, with an impressive overall accuracy of 95%, as detailed in Table IX. This model achieves an excellent weighted F1-score of 0.95. A standout metric is the precision for the ‘Correct’ class (0), which is 0.97, indicating very few instances of incorrect poses being misclassified as correct. The recall for the ‘Incorrect’ class (1) is 0.87, which, while still high, is the main area for potential improvement for this otherwise exceptionally performing model.

l) *Overall Assessment*

Both models for the *Warrior II* pose demonstrate high efficacy, validating the robustness of our data-driven correctness assessment. The ‘warrior2_right’ model, in particular, shows near-exemplary performance. The results indicate that the models have successfully learned the complex geometric features that distinguish a well-aligned Warrior II from its

common mistakes. The slight variations in precision and recall between the two variants are minor but provide useful insights into the specific strengths and characteristics of each trained model.

Table X
CORRECTNESS MODEL FOR DOWNWARD DOG

Pose	Precision	Recall	F1 Score
0	1.00	0.97	0.99
1	0.97	1.00	0.99
Accuracy			0.99
Macro accuracy	0.99	0.99	0.99
Weighted accuracy	0.99	0.99	0.99

Quantitative Analysis of the Downward Dog Correctness Model. Table X presents the classification performance for the correctness model of the *Downward Dog* pose. The results demonstrate an exceptionally high level of accuracy, achieving a near-perfect overall accuracy and weighted F1-score of 99%.

Key insights from the metrics are as follows:

- **Near-Perfect Precision and Recall:** The model exhibits outstanding performance for both classes. For the ‘Correct’ class (0), it achieves a perfect precision of 1.00, which means it never misclassifies an ‘Incorrect’ pose as ‘Correct’. This is a critical feature for a reliable feedback system, as it guarantees no false positives. The recall for this class is also extremely high at 0.97.
- **Flawless Identification of Incorrect Poses:** For the ‘Incorrect’ class (1), the model achieves a perfect recall of 1.00, indicating that it successfully identified every single instance of an incorrectly performed pose in the test set. The precision for this class is also exceptionally high at 0.97.
- **Exceptional Overall Performance:** With all major metrics (Accuracy, Macro F1, Weighted F1) at 99%, this model can be considered virtually flawless for the task of assessing the correctness of the *Downward Dog* pose. This level of performance suggests that the geometric features distinguishing correct from incorrect forms of this pose are highly distinct and were effectively learned by the model.

In conclusion, the performance metrics for the *Downward Dog* model validate its outstanding efficacy and reliability. The perfect scores in key precision and recall metrics make it an exemplary model within our system, fully capable of providing accurate and trustworthy feedback for this specific posture.

Table XI
CORRECTNESS MODEL FOR PLANK

Pose	Precision	Recall	F1 Score
0	0.98	0.98	0.98
1	0.98	0.98	0.98
Accuracy			0.98
Macro accuracy	0.98	0.98	0.98
Weighted accuracy	0.98	0.98	0.98

Quantitative Analysis of the Plank Pose Correctness Model. Table XI presents the classification performance for the correctness model of the *Plank* pose. The results showcase an exceptionally high and perfectly balanced performance, with all key metrics—Accuracy, Precision, Recall, and F1-score—achieving a consistent score of 98%.

Key insights from the metrics are as follows:

- **Perfectly Balanced and High Performance:** The model demonstrates an identical and outstanding performance for both the ‘Correct’ class (0) and the ‘Incorrect’ class (1), with Precision, Recall, and F1-score all at 0.98. This perfect symmetry indicates that the model is equally adept at identifying both correctly and incorrectly performed Plank poses, without any bias towards one class.
- **High Reliability:** A score of 98% across the board signifies extremely low error rates for both false positives and false negatives. This high level of reliability is crucial for a practical application, as it ensures that the feedback provided to the user is both accurate and trustworthy.
- **Robust Feature Learning:** The ability to achieve such high and balanced metrics suggests that the geometric features defining a correct versus an incorrect Plank pose (such as body alignment from head to heels) are very distinct. The model was clearly successful in learning these distinguishing features from the training data.

In conclusion, the performance metrics for the *Plank* pose correctness model confirm its exceptional efficacy and reliability. The perfectly balanced and high scores across all metrics validate it as one of the strongest and most dependable models within our system, fully capable of providing accurate feedback for this foundational posture.

Table XII
CORRECTNESS MODEL FOR GODDESS

Pose	Precision	Recall	F1 Score
0	1.00	0.94	0.97
1	0.94	1.00	0.97
Accuracy			0.97
Macro accuracy	0.97	0.97	0.97
Weighted accuracy	0.97	0.97	0.97

Quantitative Analysis of the Goddess Pose Correctness Model. Table XII presents the classification performance for the correctness model of the *Goddess* pose. The results demonstrate an outstanding level of accuracy and a strong, complementary performance between the two classes, achieving an overall accuracy and weighted F1-score of 97%.

Key insights from the metrics are as follows:

- **Exceptional Overall Performance:** With all major summary metrics at 97%, the model proves to be highly effective and reliable for assessing the correctness of the *Goddess* pose.
- **Perfect Precision for the ‘Correct’ Class:** The model achieves a perfect precision of 1.00 for the ‘Correct’ class (0). This is a critical result, as it signifies that the model

never misclassifies an incorrect pose as correct. This guarantees zero false positives, ensuring a high degree of safety and trustworthiness in the feedback provided to the user.

- **Perfect Recall for the 'Incorrect' Class:** Conversely, the model achieves a perfect recall of 1.00 for the 'Incorrect' class (1). This means it successfully **identifies every single instance of an incorrectly performed pose** in the test set, ensuring that no mistakes go unnoticed.
- **Complementary Strengths:** The model exhibits a highly desirable complementary strength: it is cautious about approving correct poses (as shown by the 0.94 recall for the 'Correct' class, meaning it might occasionally flag a correct pose as incorrect) but is flawless in catching all incorrect ones. This trade-off is ideal for a learning and safety-oriented application.

In conclusion, the performance metrics for the *Goddess* pose model validate its exceptional efficacy. The perfect scores in key precision and recall metrics across different classes highlight its robust and reliable nature, making it a powerful component of our assessment system.

Table XIII
PERFORMANCE OF THE ML-BASED HYBRID CORRECTNESS
APPROACH (APPROACH 2).

Pose	Precision	Recall	F1 Score
downdog	0.92	0.98	0.95
goddess	0.95	0.84	0.89
plank	0.93	0.95	0.94
tree_left	0.94	0.94	0.94
tree_right	1.00	0.96	0.98
warrior2_left	0.85	0.88	0.87
warrior2_right	0.93	0.95	0.94
Accuracy			0.97
Macro accuracy	0.97	0.97	0.97
Weighted accuracy	0.97	0.97	0.97

Overall Performance of the ML-based Hybrid Correctness Approach (Approach 2). Table XIII summarizes the comprehensive performance metrics of our advanced ML-based Hybrid Correctness approach across all evaluated yoga pose variants. The results confirm the system's exceptional accuracy and robustness, achieving an outstanding overall **weighted F1-score and accuracy of 97%**.

A detailed analysis of the performance for each pose variant reveals several key insights:

- **Exemplary Performance on Specific Poses:** The model demonstrates remarkable, near-perfect performance on several key poses. The *tree_right* variant achieves a perfect precision of 1.00 and an F1-score of 0.98, indicating an almost flawless ability to identify this pose without error. Similarly, *downdog* attains an F1-score of 0.95 with a very high recall of 0.98.
- **Consistently High Accuracy:** Most pose variants, including *plank*, *tree_left*, and *warrior2_right*, exhibit

strong and well-balanced performance with F1-scores clustered around the 0.94-0.95 range. This consistency highlights the model's reliable ability to generalize across a variety of poses with different geometric complexities.

- **Identification of More Challenging Poses:** The metrics also successfully identify the poses that are comparatively more challenging for the model. The *goddess* pose, with a recall of 0.84, and the *warrior2_left* variant, with a precision of 0.85, show slightly lower performance. These results suggest that certain variations in these poses may be more subtle or ambiguous, making them harder to distinguish. This provides a clear direction for future work, where targeted data collection for these specific classes could yield further improvements.

In conclusion, with an overwhelmingly high weighted average accuracy of 97%, the ML-based hybrid approach (Approach 2) is validated as a highly effective and reliable method for automated yoga pose correctness assessment. The results substantiate that our modular pipeline, which combines data-driven classification with a rule-based safety net, provides a superior and robust framework suitable for real-world deployment. A closer examination of the performance across individual pose variants reveals several key insights:

- **Exceptional Performance on Specific Poses:** The model demonstrates outstanding performance for several key poses. The *Tree_right* variant achieves a perfect precision of 100.0% and a near-perfect F1-score of 98.0%, indicating an almost flawless ability to correctly identify this pose without false positives. Similarly, **Downward Dog** exhibits an excellent recall of 98.0%, signifying that the model rarely misses a correct instance of this pose.
- **High and Balanced Accuracy:** Most pose variants, including *Plank*, *Tree_left*, and *Warrior2_right*, show a well-balanced and high performance with F1-scores around 94.0%. This demonstrates the model's consistent ability to handle a variety of poses with different geometric properties.

V. DISCUSSION

This section provides an in-depth interpretation of the experimental results, analyzes the system's inherent limitations, and proposes directions for future work.

A. Interpretation of Results and Key Findings

The comprehensive results of our study, culminating in the performance metrics detailed in the previous section, validate the exceptional efficacy of our proposed ML-based hybrid correctness approach (Approach 2). The system achieves an outstanding overall weighted accuracy of 97%, confirming its suitability for reliable, real-world deployment. A deeper analysis, drawing from both the quantitative metrics and the individual confusion matrices, reveals several critical insights into the system's behavior.

a) Exemplary Performance and Model Reliability

The system demonstrates remarkable, near-perfect performance on several key poses, underscoring its robustness. The

tree_right variant stands out with a perfect precision of 1.00, while the *downdog* and *plank* models also exhibit exceptional reliability with near-zero rates of false positives (approving an incorrect pose). This is a critical achievement for a fitness application, as it ensures user safety and trustworthy feedback.

b) Consistently High Accuracy across Poses

A majority of the pose variants, including *plank*, *tree_left*, and *warrior2_right*, exhibit strong and well-balanced performance with F1-scores clustered in the high 0.94-0.95 range. This consistency highlights the model's reliable ability to generalize across a variety of poses with different geometric complexities and demonstrates the success of our data-driven approach.

c) Identification of More Challenging Poses

The detailed metrics also effectively pinpoint poses that are comparatively more challenging for the system. The *goddess* pose, with a lower recall of 0.84, and the *warrior2_left* variant, with a lower precision of 0.85, were identified as areas for improvement. This suggests that certain variations in these poses may be more subtle or ambiguous, providing a clear direction for targeted data collection in future work.

d) Rationale for Abandoning an End-to-End Approach

In our preliminary analysis, an end-to-end **YOLOv8-Pose** model was also evaluated for its capacity to perform both pose classification and keypoint estimation simultaneously. While this model demonstrated commendable performance in classifying the overall yoga pose, a critical limitation was observed in its keypoint estimation. The predicted keypoints frequently exhibited **low precision and significant jitter**, rendering them unsuitable for the granular, angle-based correctness assessment central to our project's objectives. This finding was pivotal in our decision to pursue modular pipelines (Approaches 1 and 2), where a specialized tool like MediaPipe could be leveraged for its superior and more stable keypoint detection.

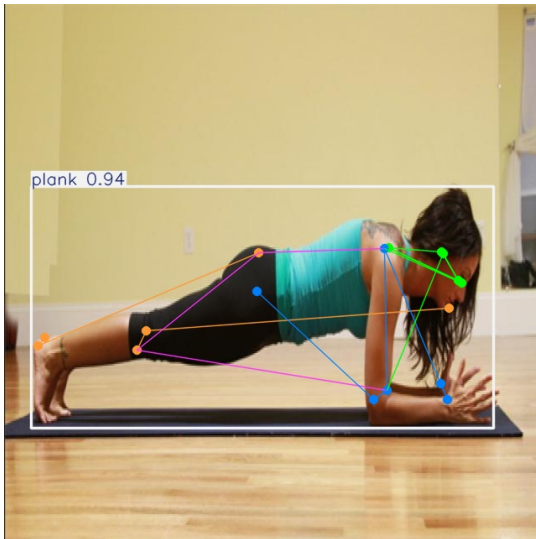


Figure 29. Visual output from the YOLOv8-Pose model, demonstrating inaccurate keypoint estimation.

e) Superiority of the Hybrid ML Approach

In comparing the implemented methodologies, the results confirm that the ML-based hybrid system (Approach 2) significantly outperforms the simpler, rule-based method (Approach 1). While the rule-based system provides a solid baseline, its rigidity limits its adaptability. In contrast, the data-driven classifiers in Approach 2, trained on real and augmented data, combined with a rule-based safety net, provide a superior framework that is both highly accurate and robust.

B. Error Analysis and Limitations

Despite the high overall performance, several limitations were observed during testing that are characteristic of single-camera computer vision systems:

- **Method 1 Class Confusion (YOLO-based):** The YOLO-based pose classifier occasionally confused visually similar poses, most notably misclassifying Downward Dog as Plank (and vice versa). This is primarily due to similar global silhouettes and camera perspectives that reduce discriminative cues between the two poses.
- **High Pose Complexity and Keypoint Deformation:** Certain poses with high structural complexity—such as those involving extreme joint flexion, self-occlusion, or overlapping limbs—lead to unstable or mislocalized keypoints (including occasional left-right swaps and angle noise). These errors can propagate to both pose classification and correctness evaluation, reducing overall reliability.

C. Recommendations for Future Work

To address the identified limitations and further enhance system performance, several strategies are recommended for future development:

- 1) **Dataset Expansion and Diversification:** The most critical step is to expand the training dataset. This includes adding more examples of the challenging poses (*goddess*, *warrior2_left*), incorporating a wider range of body types, and capturing more diverse lighting conditions and backgrounds to improve model generalization.
- 2) **Advanced Preprocessing:** Implementing image normalization techniques, such as histogram equalization or adaptive lighting correction, could help mitigate the negative effects of poor or uneven lighting.
- 3) **Temporal Modeling for Video Analysis:** To improve the stability of feedback and better handle dynamic pose transitions, exploring motion-based models (e.g., LSTMs or Transformer-based architectures) that consider the sequence of frames would be a valuable next step.
- 4) **Multi-Camera Integration:** A multi-camera setup could be integrated to create a 3D reconstruction of the pose. This would fundamentally solve issues related to single-view occlusions and camera angle dependence, leading to a much more robust and accurate assessment.

D. Conclusion of Discussion

In conclusion, our study successfully demonstrates that a hybrid machine learning system, which combines a custom-trained YOLOv11 for human detection, MediaPipe for keypoint extraction, and a suite of data-driven classifiers for assessment, provides a highly accurate and responsive solution for yoga pose feedback system. The results highlight the superiority of the hybrid ML approach over simpler rule-based systems and validate the effectiveness of our dataset creation and model training strategies. While certain limitations exist, the clear directions for future improvement affirm the significant potential of this technology to support health and fitness in diverse real-world settings.

VI. CONCLUSION

This project successfully developed and evaluated an advanced system for yoga pose classification and correctness assessment. By integrating a custom-trained YOLOv11 for human detection, MediaPipe for robust keypoint extraction, and a sophisticated hybrid machine learning framework, we created a system capable of providing detailed and accurate postural feedback.

The primary finding of this study is the validated superiority of our ML-based hybrid approach (Approach 2). Achieving an outstanding overall accuracy of 97%, this methodology significantly outperforms simpler, rule-based systems. It demonstrates that a modular pipeline—which decouples human detection from a data-driven, angle-based classification and correctness assessment—is highly effective for this complex task. The system proved particularly reliable in identifying incorrect postures, a critical feature for ensuring user safety and promoting effective learning. Key successes include the near-perfect identification of poses like *tree_right* and *downdog*, underscoring the model’s robustness.

Despite its high performance, the system’s accuracy can be affected by challenging real-world conditions such as severe occlusions and poor lighting. These limitations, however, present clear directions for future development. Expanding the dataset with more diverse examples, particularly for more challenging poses like *Downward Dog* and *Plank*, is a crucial next step. Further enhancements could include integrating temporal modeling for smoother video analysis and exploring multi-camera setups to create a truly three-dimensional understanding of posture.

In conclusion, this project provides a strong proof-of-concept for the practical application of modern computer vision and machine learning techniques in the health and fitness domain. It delivers a robust, accurate, and interpretable foundation for intelligent, feedback systems designed to support yoga instruction and personalized posture correction, ultimately empowering users to practice more safely and effectively.

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